Topics in Chapter 1

1.1 Introduction
1.2 Classes of computers
1.3 Defining computer architecture and What's the task of computer design?
1.4 Trends in Technology
1.5 Trends in power in Integrated circuits
1.6 Trends in Cost
1.7 Dependability
1.8 Measuring, Reporting and summerizing Perf.
1.9 Quantitative Principles of computer Design
1.10 Putting it altogether

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- What's CA?
- What's a CA designer's task ?

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What's Computer Architecture

- Patterson's talk : "50 years of Computer Architecture"
  - https://www.bilibili.com/video/av46710093/
- The future of computing- a conversation with John Hennessy
  - https://www.bilibili.com/video/av23283946/
- **Technology**
  - A key factor in the long run of CPU performance improvement.

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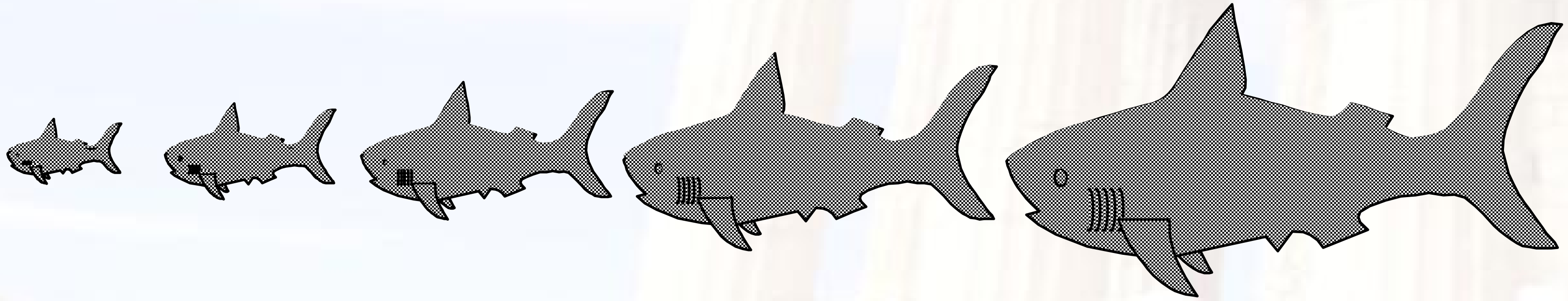
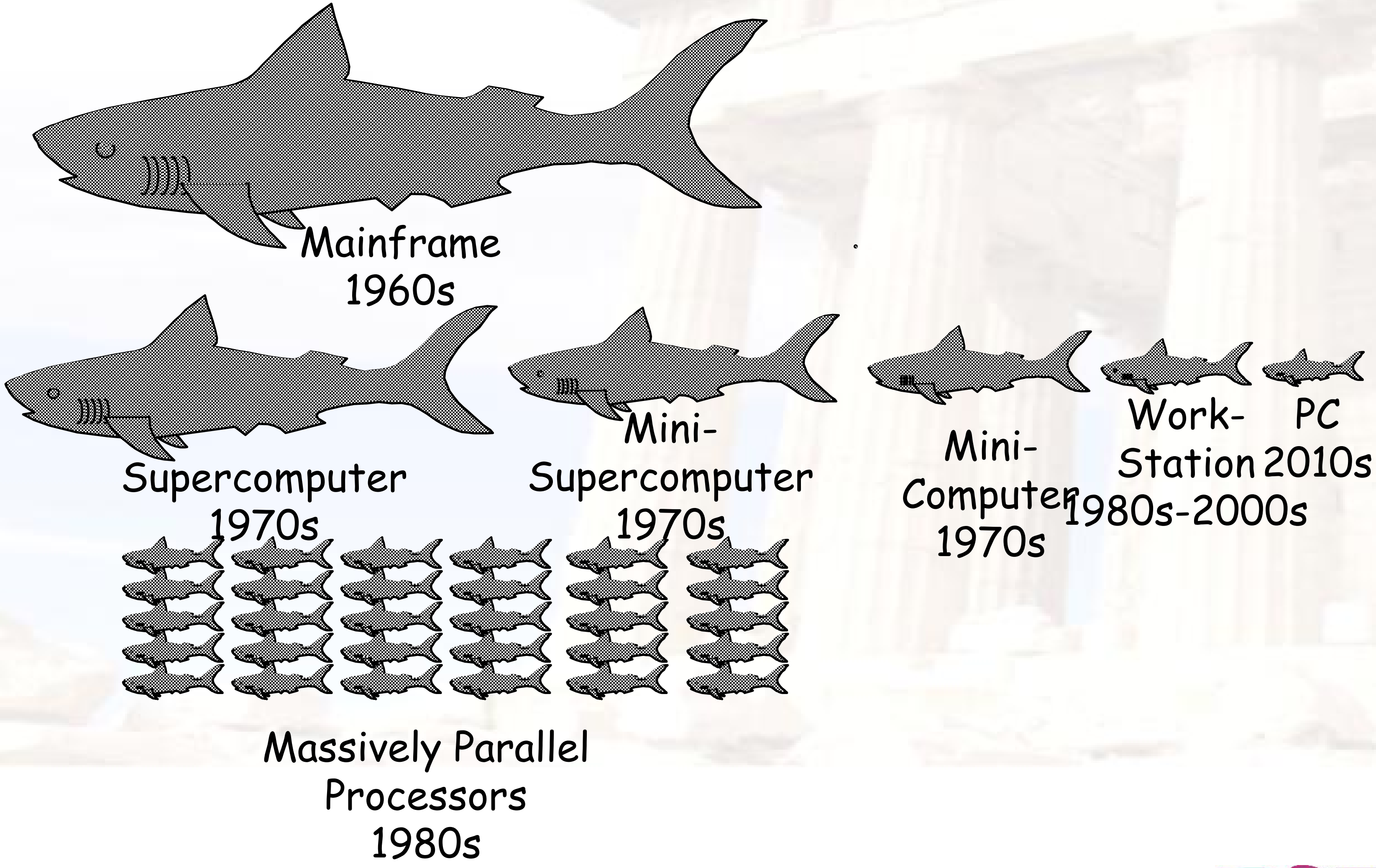
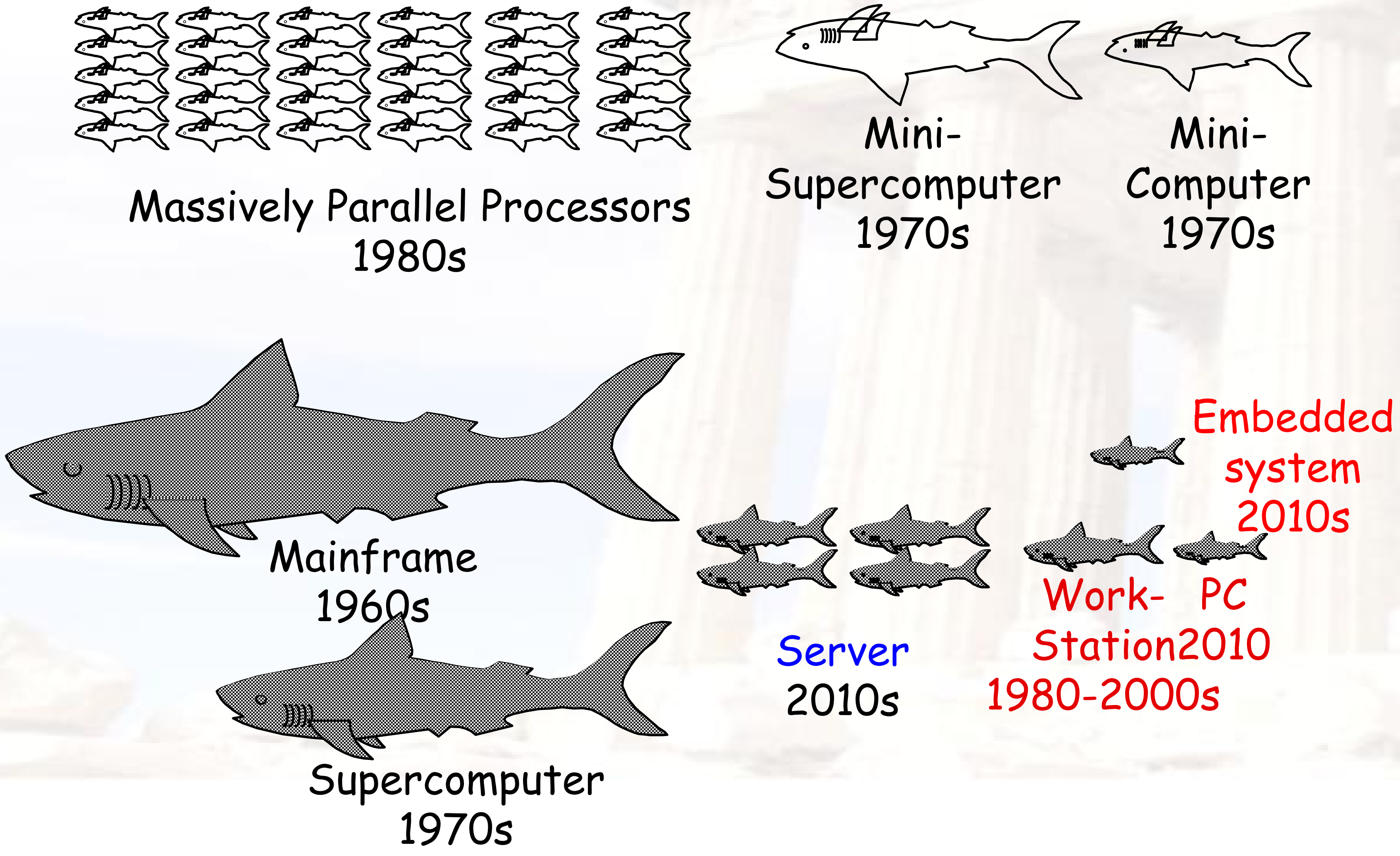
- Original:

Big Fishes Eating Little Fishes




Mainframe
1960s
Supercomputer
1970s
Mini-
Supercomputer
1970s
Mini-
Computer
1970s
Work-
Station
1980s-2000s
PC
2010s
Massively Parallel
Processors
1980s



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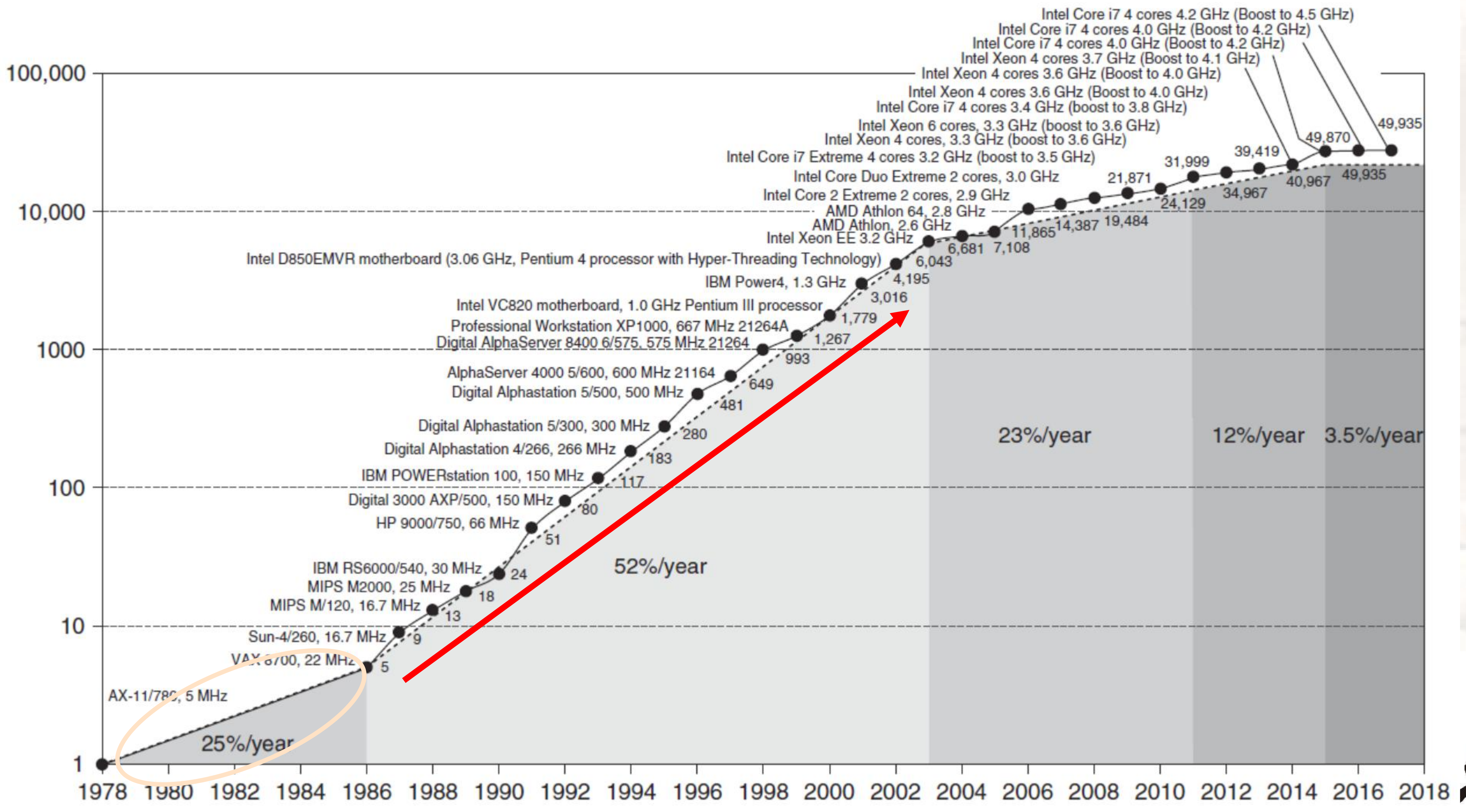
Incredible CPU performance improvement

VAX : 25%/year 1978 to 1986； RISC+x86: 52%/year 1986 to 2002； RISC+x86: 22%/year after 2002



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What the figure tell us ?

- Performance improvements:

  25%: Technological improvements more steady than progress in computer architecture.

  - Feature size, clock speed

- 52%: After RISC emergence, computer design emphasized both architectural innovation and efficient use of technology improvements.
  - Computer Architecture plays an important role in performance improvement
  - **Pipeline, dynamic scheduling, ooo, branch prediction, speculation, superscalar, VLIW, prediction instructions, ……**

- Technology Advances
  - CMOS VLSI dominates older technologies (TTL, ECL) in cost AND performance
- Computer architecture advances improves low-end
  - RISC, superscalar, OOO, Speculation, VLIW, RAID, …
- Together have enabled:
  - Lightweight computers
  - Productivity-based managed/interpreted programming languages
- Price: Lower costs due to …
  - Simpler development: smaller systems, fewer components
  - Higher volumes: same dev. cost 10,000 vs. 10,000,000 units
  - Lower margins by class of computer, due to fewer services
- Function and Usage
  - scientific computing →non-digital processing → embedded system
  - Rise of networking/local interconnection technology

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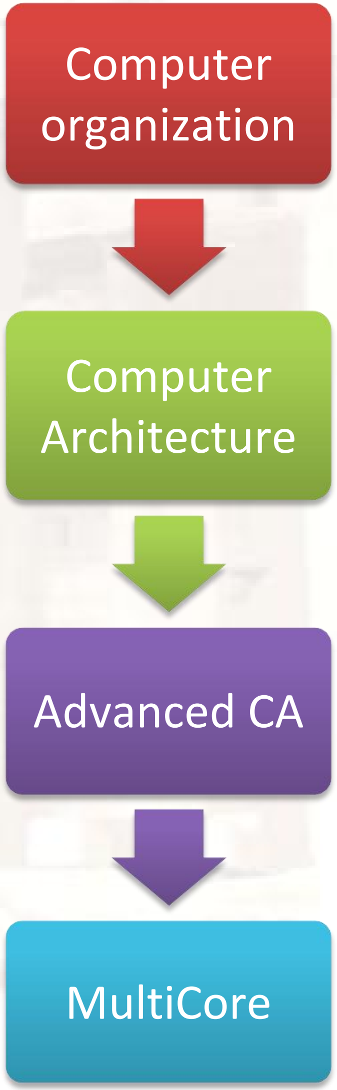
- The Decade of the 1970's "Microprocessors"
  - Programmable Controller (microprogramming)
  - Single-Chip Microprocessors
  - Personal Computers (PC)
- The Decade of the 1980's "Quantitative Architecture"
  - Instruction Pipelining
  - Fast Cache Memories
  - Compiler Considerations
  - Workstations
- The Decade of the 1990's "Instruction-Level Parallelism"
  - Superscalar Processors
  - Speculative Microarchitectures
  - Aggressive Code Scheduling
  - Low-Cost Desktop/Supercomputing
- The Decade of the 2000's "Thread-level/Data-level parallelism"



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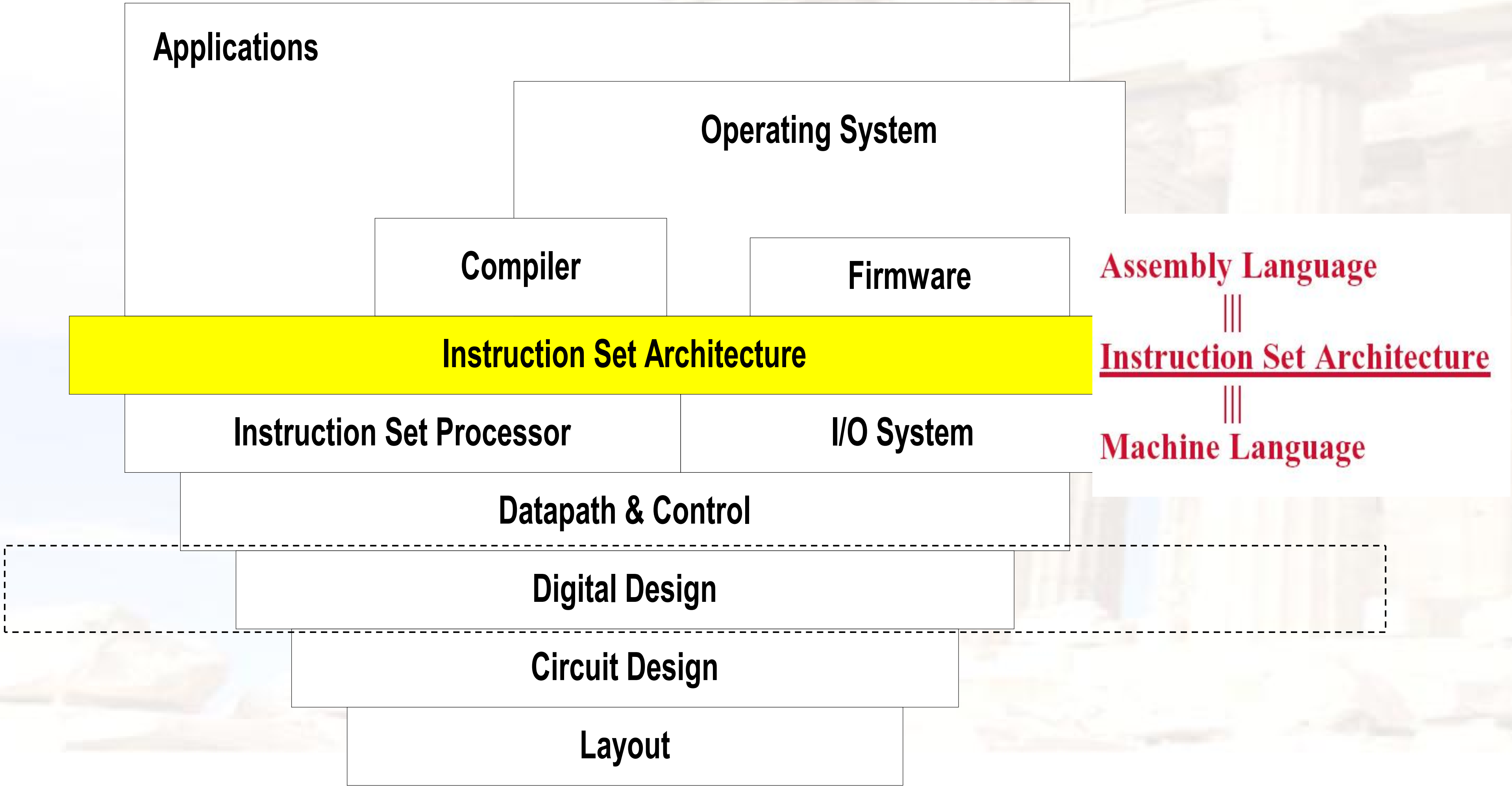
What's Computer Architecture

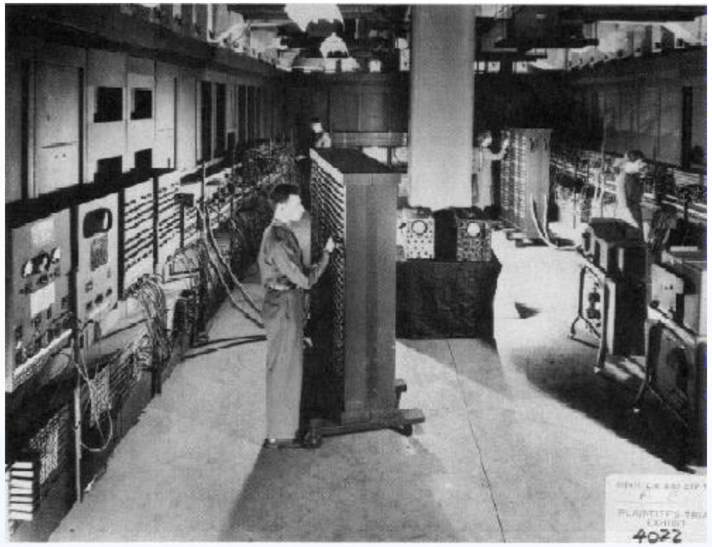
❑Concept Evolution

The attributes of a [computing] system as seen by the programmer, i.e., the conceptual structure and functional behavior, as distinct from the organization of the data flows and controls the logic design, and the physical implementation.

Amdahl, Blaaw, and Brooks, 1964


Applications
Operating System
Compiler
Firmware
Instruction Set Architecture
Instruction Set Processor
I/O System
Datapath & Control
Digital Design
Circuit Design
Layout
Assembly Language
Instruction Set Architecture
Machine Language

Very different appearance



From Computer Desktop Encyclopedia
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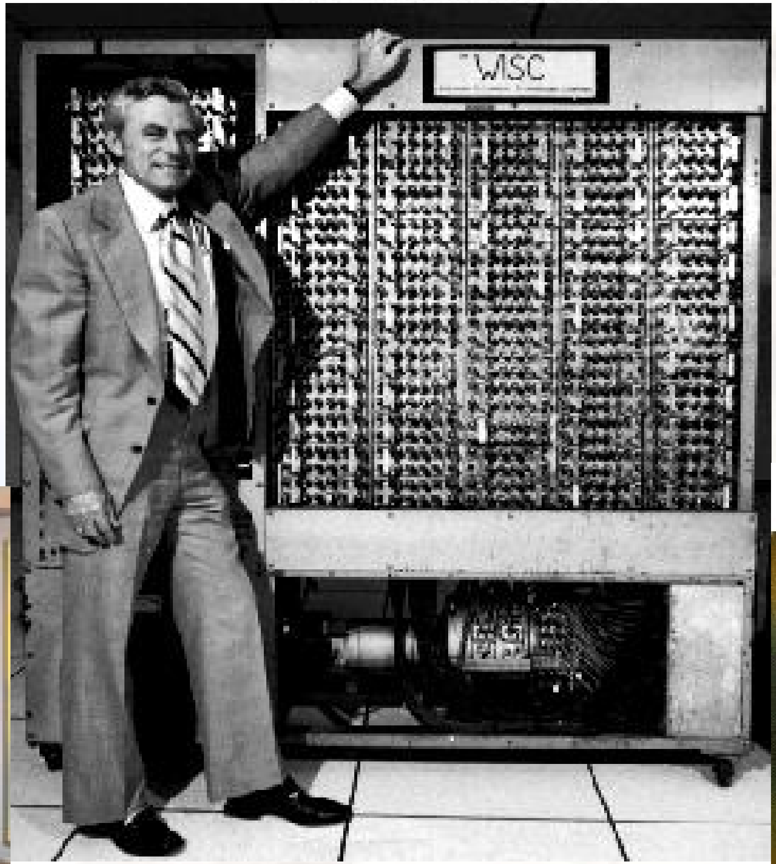

From Computer Desktop Encyclopedia
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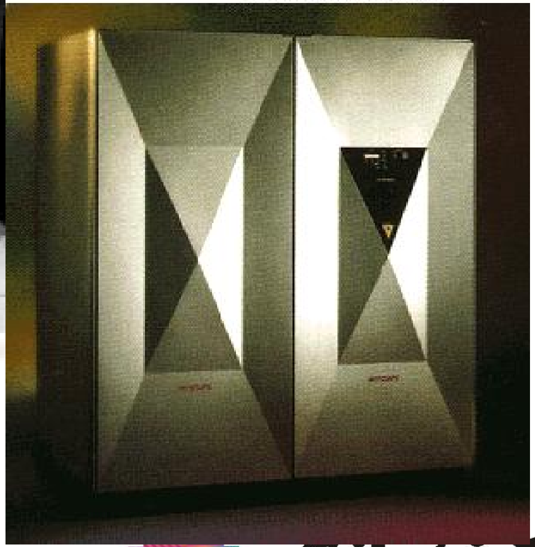
© 1997 Amdahl Corporation



Very Different ISA

- PDP-11
- IBM 360
- VAX
- CRAY
- …

- LC3
- 80x86
- PowerPC
- MIPS
- ARM
- RISC V

New Concepts ?

Tradeoff

- *Computer Architecture* is the science and art of selecting and interconnecting hardware components to create computers that meet functional, performance, cost and power goals.

- It is a **blueprint** and functional description of requirements and design implementations for the various parts of a computer, focusing largely on the way by which the central processing unit (CPU) performs internally and accesses addresses in memory.

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Computer architecture

- Computer architecture comprises at least three main subcategories:[1]
- ***Instruction set architecture***,
- ***Microarchitecture***, also known as ***Computer organization*** is a lower level, more concrete and detailed, description of the system that involves how the constituent parts of the system are interconnected and how they interoperate in order to implement the ISA.[
- ***System Design*** which includes all of the other hardware components within a computing system such as:
  - Logic Implementation
  - Circuit Implementation
  - Physical Implementation

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Seven dimensions of ISA

- Class of ISA
- Memory addressing
- Addressing modes
- Types and sizes of operands
- Operations
- Control flow instructions
- Encoding an ISA

Computer Applications

- **Architects need to understand applications' behavior**
  - **We say we design general purpose processors, but they really focus on specific sets of applications**
  - **Architecture can be tuned to applications**
- **Types of applications today**
  - **Scientific**
    - Weather prediction, crash analysis, earthquake analysis, medical imaging, imaging of the earth (searching for oil)
  - **Business**
    - database, data mining, video
  - **General purpose**
    - Microsoft Word, Excel
  - **Real-time**
    - automated control systems,
  - Others: Games, Mobile，Internet of Things，5G

The Task of Computer Design-2

- Determine the important attributes of a new machine to maximize performance while staying with constrains, such as cost, power, availability, etc.
  - instruction set architecture design
  - functional organization
    - High level aspects of computer design, i.e. memory system, bus architecture and internal CPU design.
  - logic design (hardware)
  - implementation (hardware)

Incredible CPU performance improvement

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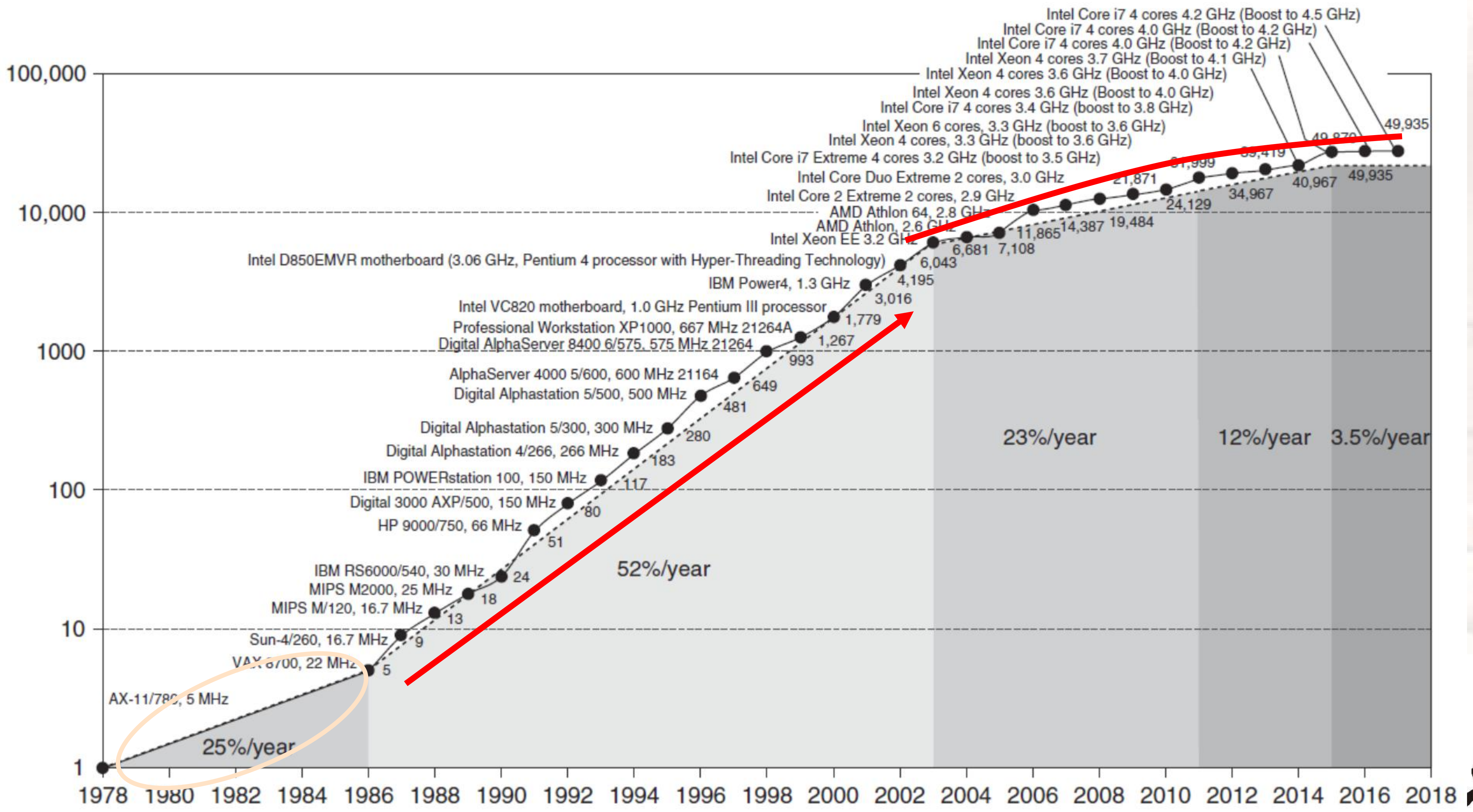

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Challenges of “three walls”

ILP Wall

- diminishing returns on finding more ILP HW (Explicit thread and data parallelism must be exploited)

Memory Wall

- growing disparity of speed between CPU and memory outside the CPU chip. Memory latency would become an overwhelming bottleneck in computer performance.

Power Wall

- the trend of consuming double the power with each doubling of operating frequency

Trends of Computer Architecture

- ILP → TLP(thread level parallel)
  → DLP(data level parallel),

  Why?

  Google file system, MapReduce, column-oriented DB

- 2D memory → 3D stack memory

- Performance → Power-aware design

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Computational RAM / PIM

- Processor in memory (PIM)
  - Processing in memory (PIM, sometimes called *processor in memory*) is the integration of a processor with RAM (random access memory) on a single chip. The result is sometimes known as a *PIM chip*.

- 1995. 4 IEEE computer

  Processing in memory: the Terasys massively Parallel PIM Array.

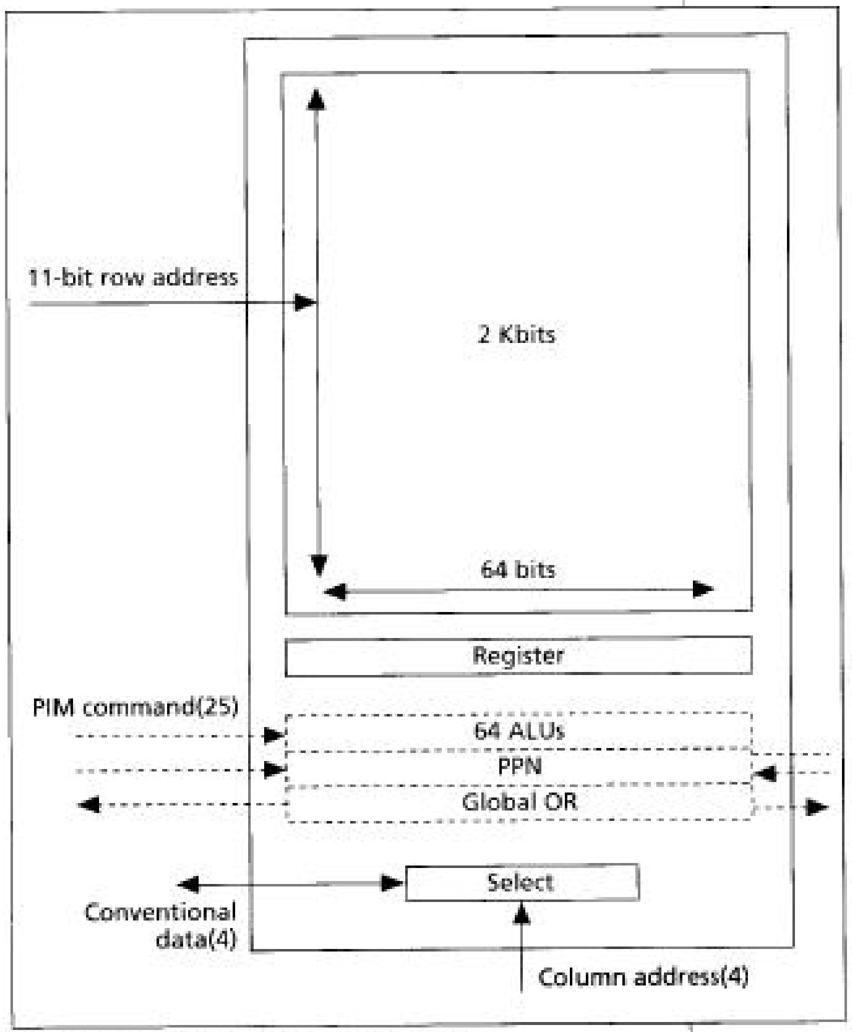

Figure 2. A processor-in-memory chip.

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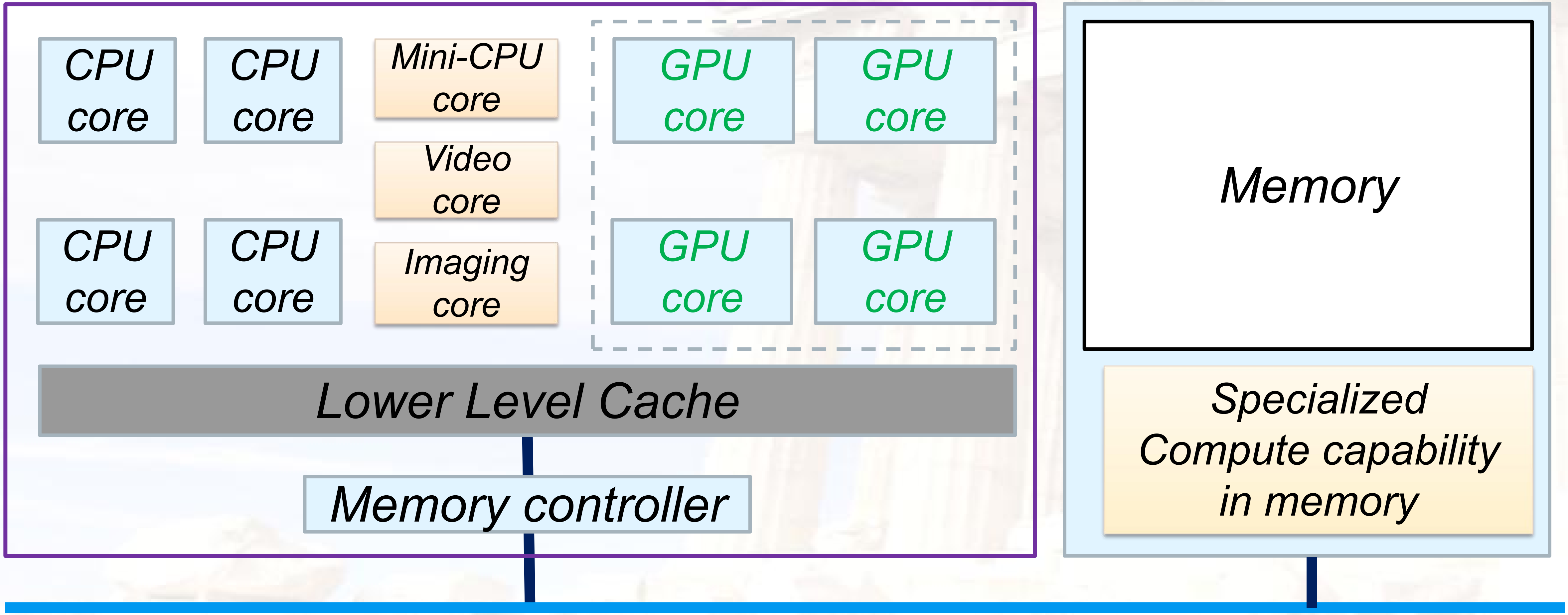
Memory as an Accelerator



Bitwise Operation： In-DRAM copy，zero，and，or，not

Current Trends in Architecture

- Cannot continue to leverage Instruction-Level parallelism (ILP)
  - Single processor performance improvement ended in 2003

- New models for performance:
  - Data-level parallelism (DLP)
  - Thread-level parallelism (TLP)
  - Request-level parallelism (RLP)

- These require explicit restructuring of the application

Recent research fields

- Power-aware computer architecture
- Reconfigurable computer architecture
- Multicore, Manycore,
- HPS : P→T→ E-level computer
- PIM： Processor in Memory
- AI processor ------GPU & FPGA
- DSA： Domain Specific Architecture

- **Hardware and Software Codesign**

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Where to explore the parallelism ?

- Implicity, compiler and hardware

→ Explicity, programmer

So, YOU, programmers have to know parallelism in hardware, and to explore parallelism when design Algorithm and programming !

application

Algorithm

Language

Compiler

Hardware

Hardware → Hardware and compiler → compiler and programmer

ILP, Loop-level Parallism → TLP, DLP

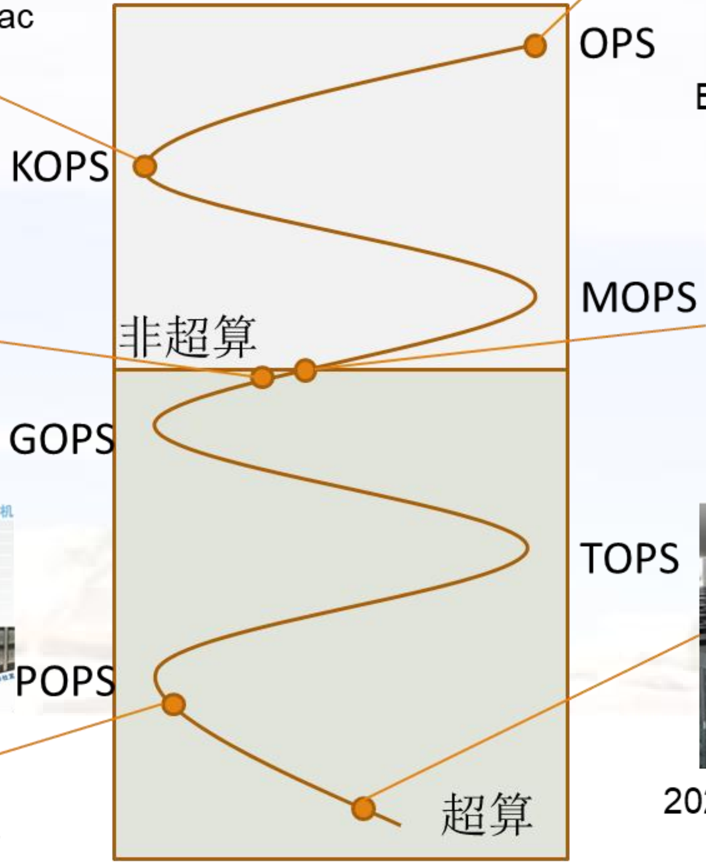
❑Something more about CA

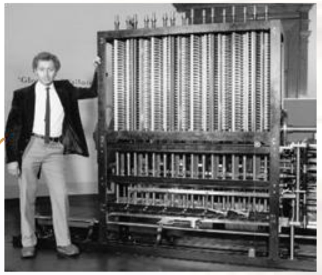
Fastest computer

1949
Edsac

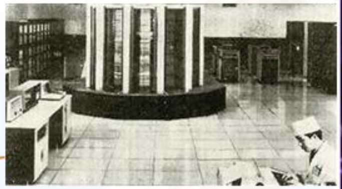
1976 Cray 1
250MOPS
世界首台超算

2009 天河 1
2.57POPS
中国超算首次领跑

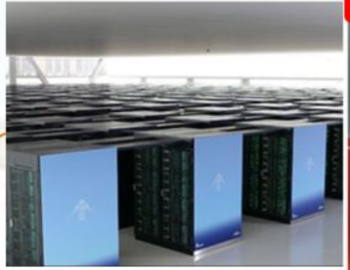




1823
Babbage Difference Engine

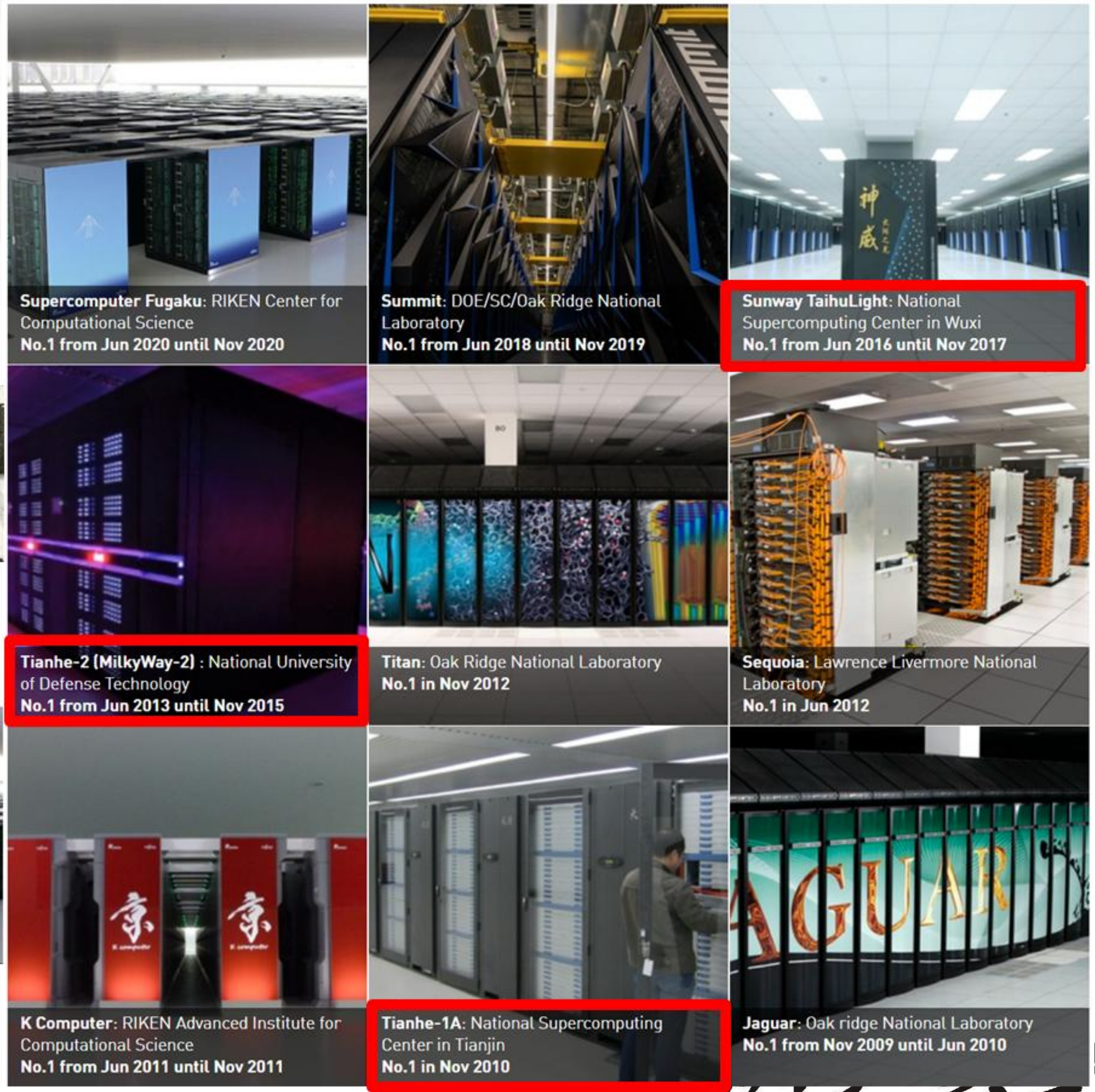


1983 银河1
100MPOS
中国首台超算



2021 Fugaku（日本）
442POPS

历年TOP1



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Fastest computer (2025.11 Top10)

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	**El Capitan** - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,340,000	1,809.00	2,821.10	29,685
2	**Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	**Aurora** - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	**JUPITER Booster** - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN EuroHPC/FZJ Germany	4,801,344	1,000.00	1,226.28	15,794
5	**Eagle** - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	
6	**HPC6** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, RHEL 8.9, HPE Eni S.p.A. Italy	3,143,520	477.90	606.97	8,461
7	**Supercomputer Fugaku** - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
8	**Alps** - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE Swiss National Supercomputing Centre (CSCS) Switzerland	2,121,600	434.90	574.84	7,124
9	**LUMI** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	531.51	7,107
10	**Leonardo** - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, EVIDEN EuroHPC/CINECA Italy	1,824,768	241.20	306.31	7,494

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Fastest computer in China

- 2011 Tianhe-1A - MPP
- 2008 Dawning 5000A
  - 30720 node * AMD Opteron 1.9Ghz QC
  - Memory: 122.88TB, Infiniband, 180.6 TeraFLOPS
  - OS: Windows HPC 2008
  - Rank 10 in top 500 in Nov. 2008
- 2004 Dawning 4000A
  - 11 TeraFLOPS
  - rank 10 in top 500 in June, 2004
- 2003 ShenTeng6800
  - 5.324 TeraFLOPS
- 2002 ShenTeng1800
  - 2.04 TeraFLOPS
- 2000   YinHe IV
  - 1024个CPU
  - 1 TeraFLOPS

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What are the fastest computers?

- http://top500.org/ June. 2011

1 K computer, SPARC64 VIIIfx 2.0GHz, Tofu interconnect

2 Tianhe-1A - MPP, X5670 2.93Ghz 6C, NVIDIA GPU, FT-1000 8C

3 Jaguar - Cray XT5-HE Opteron 6-core 2.6 GHz

4 Nebulae - Dawning TC3600 Blade, Intel X5650, NVidia Tesla C2050 GPU

5 TSUBAME 2.0 - HP ProLiant SL390s G7 Xeon 6C X5670, Nvidia GPU, Linux/Windows

6 Cielo - Cray XE6 8-core 2.4 GHz

7 Pleiades - SGI Altix ICE 8200EX/8400EX, Xeon HT QC 3.0/Xeon 5570/5670 2.93 Ghz, Infiniband

8 Hopper - Cray XE6 12-core 2.1 GHz

9 Tera-100 - Bull bullx super-node S6010/S6030

10 Roadrunner - BladeCenter QS22/LS21 Cluster, PowerXCell 8i 3.2 Ghz / Opteron DC 1.8 GHz, Voltaire Infiniband

Fastest Supercomputer in the world

http://top500.org/ June. 2013

Rank	Site	System	Cores	Rmax (TFlop/s)	Tpeak (TFlop/s)	Power (kW)
1	National University of Defense Technology China	**Tianhe-2 (MilkyWay-2)** - TH-IVB-FEP Cluster, Intel Xeon E5-2692 12C 2.200GHz, TH Express-2, Intel Xeon Phi 31S1P NUDT	3,120,000	33,862.7	54,902.4	17,808
2	DOE/SC/Oak Ridge National Laboratory USA	**Titan** - Cray XK7 , Opteron 6274 16C 2.200GHz, Cray Gemini interconnect, NVIDIA K20x Cray Inc.	560,640	17,590.0	27,112.5	8,209
3	DOE/NNSA/LLNL USA	**Sequoia** - BlueGene/Q, Power BQC 16C 1.60 GHz, Custom IBM	1,572,864	17,173.2	20,132.7	7,890
4	RIKEN Advanced Institute for Computational Science (AICS) Japan	K computer, SPARC64 VIIIfx 2.0GHz, Tofu interconnect Fujitsu	705,024	10,510.0	11,280.4	12,660
5	DOE/SC/Argonne National Laboratory USA	**Mira** - BlueGene/Q, Power BQC 16C 1.60GHz, Custom IBM	786,432	8,586.6	10,066.3	3,945
6	Texas Advanced Computing Center/Univ. of Texas USA	**Stampede** - PowerEdge C8220, Xeon E5-2680 8C 2.700GHz, Infiniband FDR, Intel Xeon Phi SE10P Dell	462,462	5,168.1	8,520.1	4,510
7	Forschungszentrum Juelich (FZJ) Germany	**JUQUEEN** - BlueGene/Q, Power BQC 16C 1.600GHz, Custom Interconnect IBM	458,752	5,008.9	5,872.0	2,301
8	DOE/NNSA/LLNL USA	**Vulcan** - BlueGene/Q, Power BQC 16C 1.600GHz, Custom Interconnect IBM	393,216	4,293.3	5,033.2	1,972
9	Leibniz Rechenzentrum Germany	**SuperMUC** - iDataPlex DX360M4, Xeon E5-2680 8C 2.70GHz, Infiniband FDR IBM	147,456	2,897.0	3,185.1	3,423
10	National Supercomputing Center in Tianjin China	**Tianhe-1A** - NUDT YH MPP, Xeon X5670 6C 2.93 GHz, NVIDIA 2050 NUDT	186,368	2,566.0	4,701.0	4,040

Fastest Supercomputer in the world

from http://top500.org/ June，2015

RANK	SITE	SYSTEM	CORES	(TFLOP/S)	(TFLOP/S)	(KW)
1	National Super Computer Center in Guangzhou China	**Tianhe-2 (MilkyWay-2)** - TH-IVB-FEP Cluster, Intel Xeon E5-2692 12C 2.200GHz, TH Express-2, Intel Xeon Phi 31S1P NUDT	3,120,000	33,862.7	54,902.4	17,808
2	DOE/SC/Oak Ridge National Laboratory United States	**Titan** - Cray XK7 , Opteron 6274 16C 2.200GHz, Cray Gemini interconnect, NVIDIA K20x Cray Inc.	560,640	17,590.0	27,112.5	8,209
3	DOE/NNSA/LLNL United States	**Sequoia** - BlueGene/Q, Power BQC 16C 1.60 GHz, Custom IBM	1,572,864	17,173.2	20,132.7	7,890
4	RIKEN Advanced Institute for Computational Science (AICS) Japan	K computer, SPARC64 VIIIfx 2.0GHz, Tofu interconnect Fujitsu	705,024	10,510.0	11,280.4	12,660
5	DOE/SC/Argonne National Laboratory United States	**Mira** - BlueGene/Q, Power BQC 16C 1.60GHz, Custom IBM	786,432	8,586.6	10,066.3	3,945
6	Swiss National Supercomputing Centre (CSCS) Switzerland	**Piz Daint** - Cray XC30, Xeon E5-2670 8C 2.600GHz, Aries interconnect , NVIDIA K20x Cray Inc.	115,984	6,271.0	7,788.9	2,325
7	King Abdullah University of Science and Technology Saudi Arabia	**Shaheen II** - Cray XC40, Xeon E5-2698v3 16C 2.3GHz, Aries interconnect Cray Inc.	196,608	5,537.0	7,235.2	2,834
8	Texas Advanced Computing Center/Univ. of Texas United States	**Stampede** - PowerEdge C8220, Xeon E5-2680 8C 2.700GHz, Infiniband FDR, Intel Xeon Phi SE10P Dell	462,462	5,168.1	8,520.1	4,510
9	Forschungszentrum Juelich (FZJ) Germany	**JUQUEEN** - BlueGene/Q, Power BQC 16C 1.600GHz, Custom Interconnect IBM	458,752	5,008.9	5,872.0	2,301
10	DOE/NNSA/LLNL	**Vulcan** - BlueGene/Q, Power BQC 16C 1.600GHz,	393,216	4,293.3	5,033.2	1,972

Fastest Supercomputer in the world

from http://top500.org/ June. 2016

Rank	Site	System	Cores	Rmax (TFlop/s)	Rpeak (TFlop/s)	Power (kW)
1	National Supercomputing Center in Wuxi China	**Sunway TaihuLight** - Sunway MPP, Sunway SW26010 260C 1.45GHz, Sunway NRCPC	10,649,600	93,014.6	125,435.9	15,371
2	National Super Computer Center in Guangzhou China	**Tianhe-2 (MilkyWay-2)** - TH-IVB-FEP Cluster, Intel Xeon E5-2692 12C 2.200GHz, TH Express-2, Intel Xeon Phi 31S1P NUDT	3,120,000	33,862.7	54,902.4	17,808
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7	DOE/NNSA/LANL/SNL United States	**Trinity** - Cray XC40, Xeon E5-2698v3 16C 2.3GHz, Aries interconnect Cray Inc.	301,056	8,100.9	11,078.9	
8	Swiss National Supercomputing Centre (CSCS) Switzerland	**Piz Daint** - Cray XC30, Xeon E5-2670 8C 2.600GHz, Aries interconnect , NVIDIA K20x Cray Inc.	115,984	6,271.0	7,788.9	2,325
9	HLRS - Höchstleistungsrechenzentrum Stuttgart Germany	**Hazel Hen** - Cray XC40, Xeon E5-2680v3 12C 2.5GHz, Aries interconnect Cray Inc.	185,088	5,640.2	7,403.5	
10	King Abdullah University of Science and Technology	**Shaheen II** - Cray XC40, Xeon E5-2698v3 16C 2.3GHz, Aries interconnect	196,608	5,537.0	7,235.2	2,834

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Fastest Supercomputer in the world

from http://top500.org/ June. 2017

Rank	System	Cores	Rmax (TFlop/s)	Rpeak (TFlop/s)	Power (kW)
1	**Sunway TaihuLight** - Sunway MPP, Sunway SW26010 260C 1.45GHz, Sunway , NRCPC National Supercomputing Center in Wuxi China	10,649,600	93,014.6	125,435.9	15,371
2	**Tianhe-2 (MilkyWay-2)** - TH-IVB-FEP Cluster, Intel Xeon E5-2692 12C 2.200GHz, TH Express-2, Intel Xeon Phi 31S1P , NUDT National Super Computer Center in Guangzhou China	3,120,000	33,862.7	54,902.4	17,808
3	**Piz Daint** - Cray XC50, Xeon E5-2690v3 12C 2.6GHz, Aries interconnect , NVIDIA Tesla P100 , Cray Inc. Swiss National Supercomputing Centre (CSCS) Switzerland	361,760	19,590.0	25,326.3	2,272
4	**Titan** - Cray XK7, Opteron 6274 16C 2.200GHz, Cray Gemini interconnect, NVIDIA K20x , Cray Inc. DOE/SC/Oak Ridge National Laboratory United States	560,640	17,590.0	27,112.5	8,209
5	**Sequoia** - BlueGene/Q, Power BQC 16C 1.60 GHz, Custom , IBM DOE/NNSA/LLNL United States	1,572,864	17,173.2	20,132.7	7,890
6	**Cori** - Cray XC40, Intel Xeon Phi 7250 68C 1.4GHz, Aries interconnect , Cray Inc. DOE/SC/LBNL/NERSC United States	622,336	14,014.7	27,880.7	3,939
7	**Oakforest-PACS** - PRIMERGY CX1640 M1, Intel Xeon Phi 7250 68C 1.4GHz, Intel Omni-Path , Fujitsu Joint Center for Advanced High Performance Computing Japan	556,104	13,554.6	24,913.5	2,719
8	K computer, SPARC64 VIIIfx 2.0GHz, Tofu interconnect , Fujitsu RIKEN Advanced Institute for Computational Science (AICS) Japan	705,024	10,510.0	11,280.4	12,660
9	**Mira** - BlueGene/Q, Power BQC 16C 1.60GHz, Custom , IBM DOE/SC/Argonne National Laboratory United States	786,432	8,586.6	10,066.3	3,945
10	**Trinity** - Cray XC40, Xeon E5-2698v3 16C 2.3GHz, Aries interconnect , Cray Inc.	301,056	8,100.9	11,078.9	4,233

Overview of TH-2

Items	Configuration
Processors	32000 Intel Xeon CPUs + 48000 Xeon Phis + 4096 FT CPUs Peak performance is 54.9 Pflops, sustained performance 33.9PFlops
Interconnect	Proprietary high speed interconnection network TH Express-2
Memory	1.4PB in total
Storage	Global shared parallel storage system, 12.4 PB
Cabinets	125+13+24+8 = 170 Compute/Communication/Storage/Service Cabinets
Power	17.8MW
Cooling	Closed Air Cooling System

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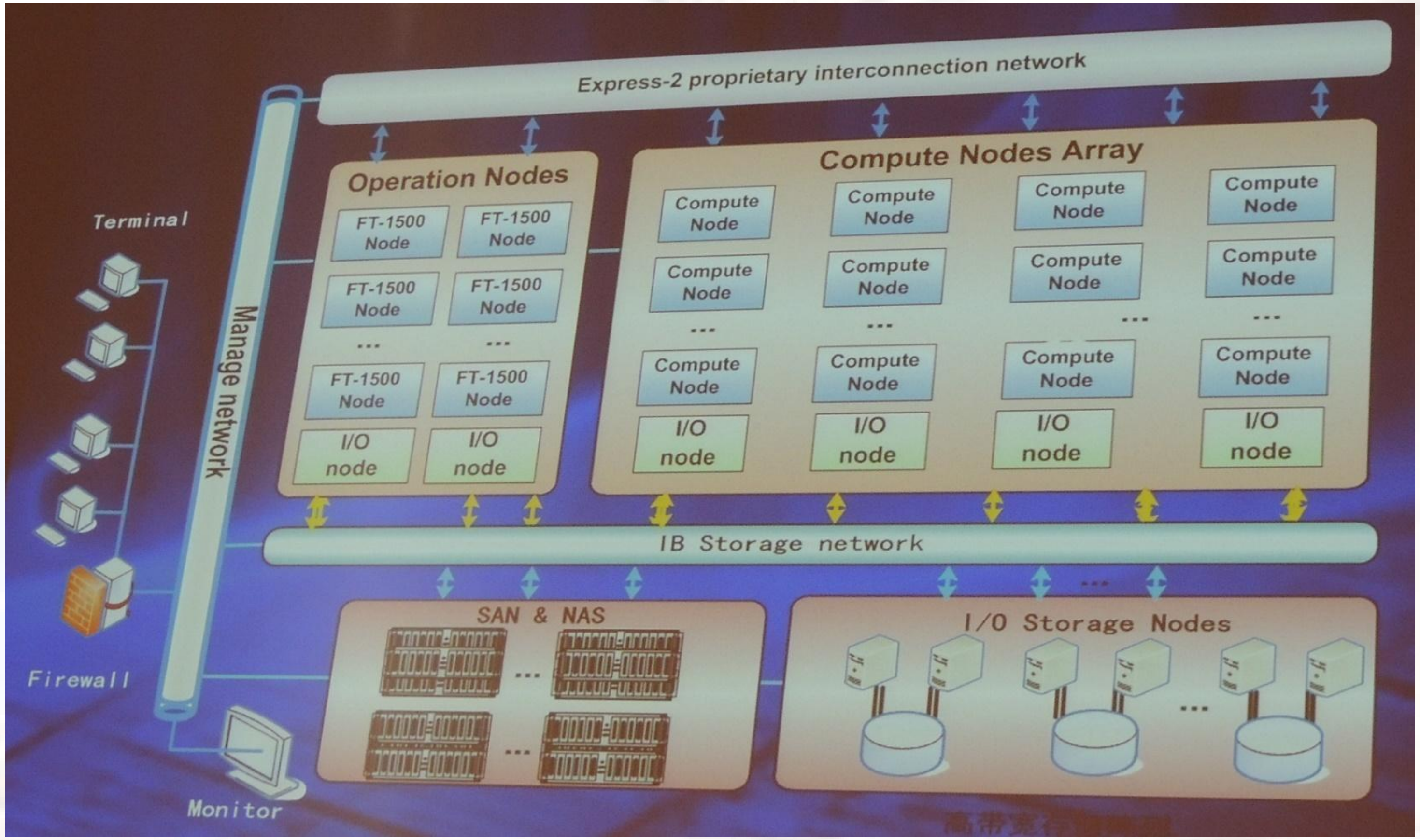
Hardware subsystem of TH-2



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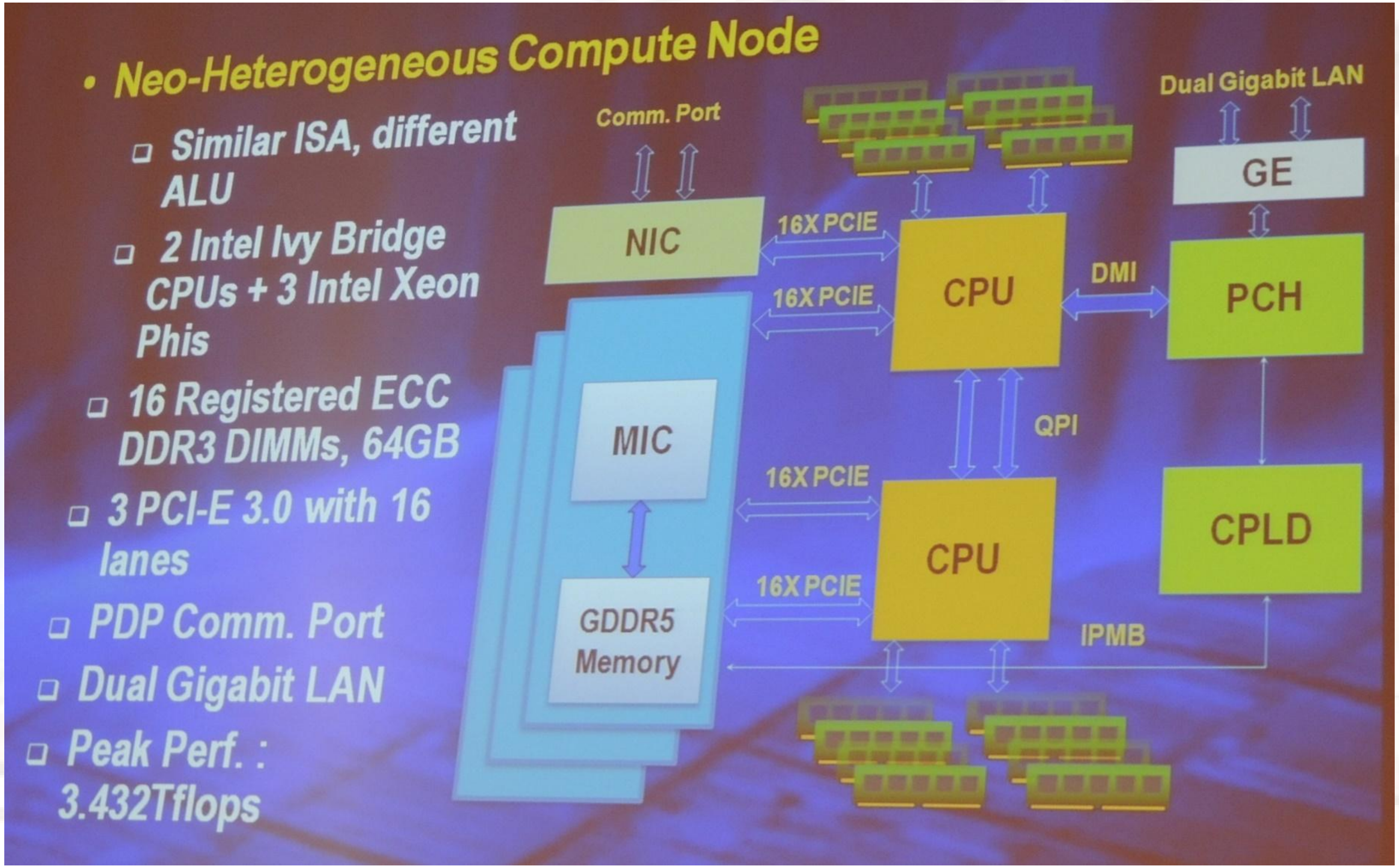
Comuter Node of TH-2



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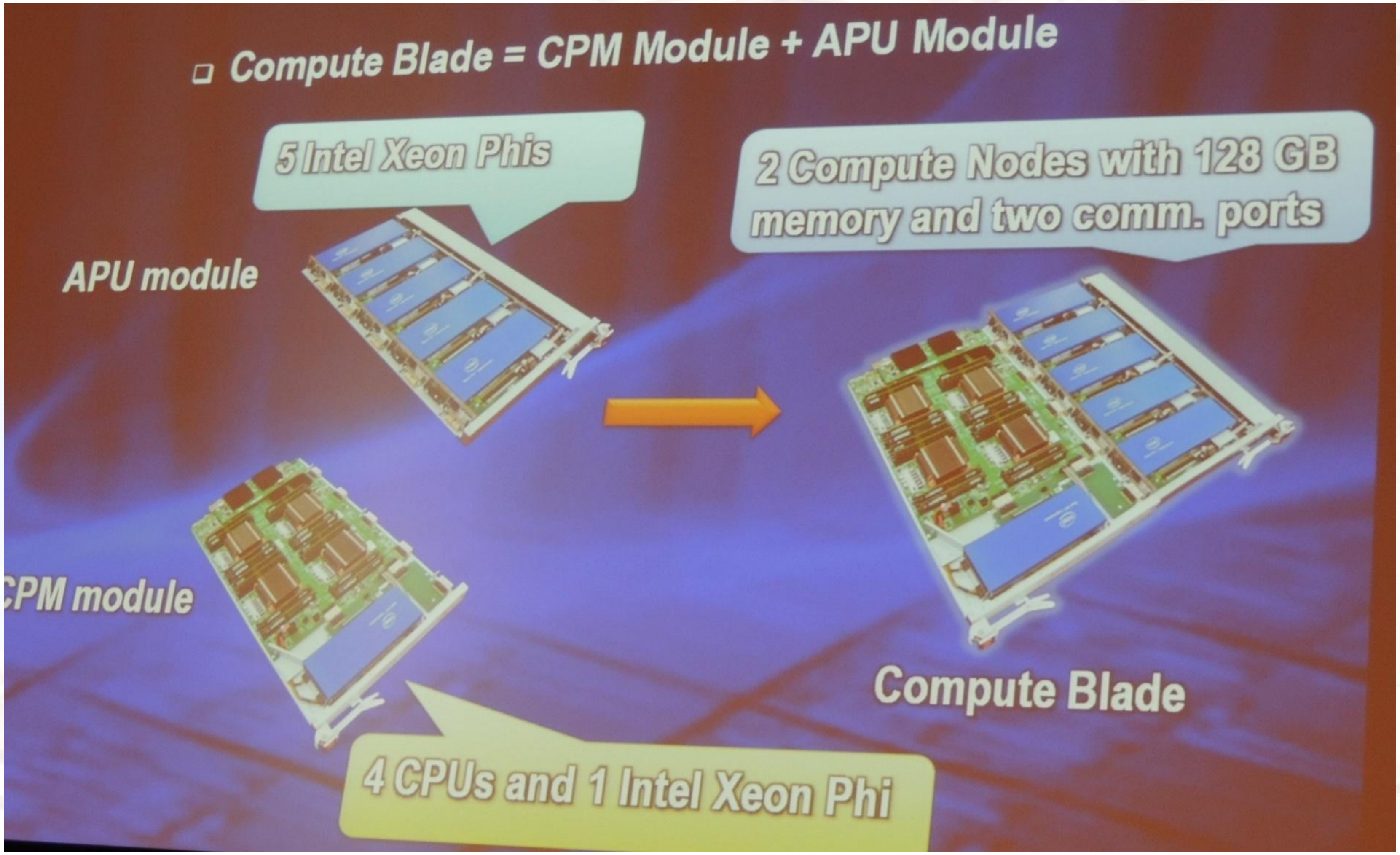
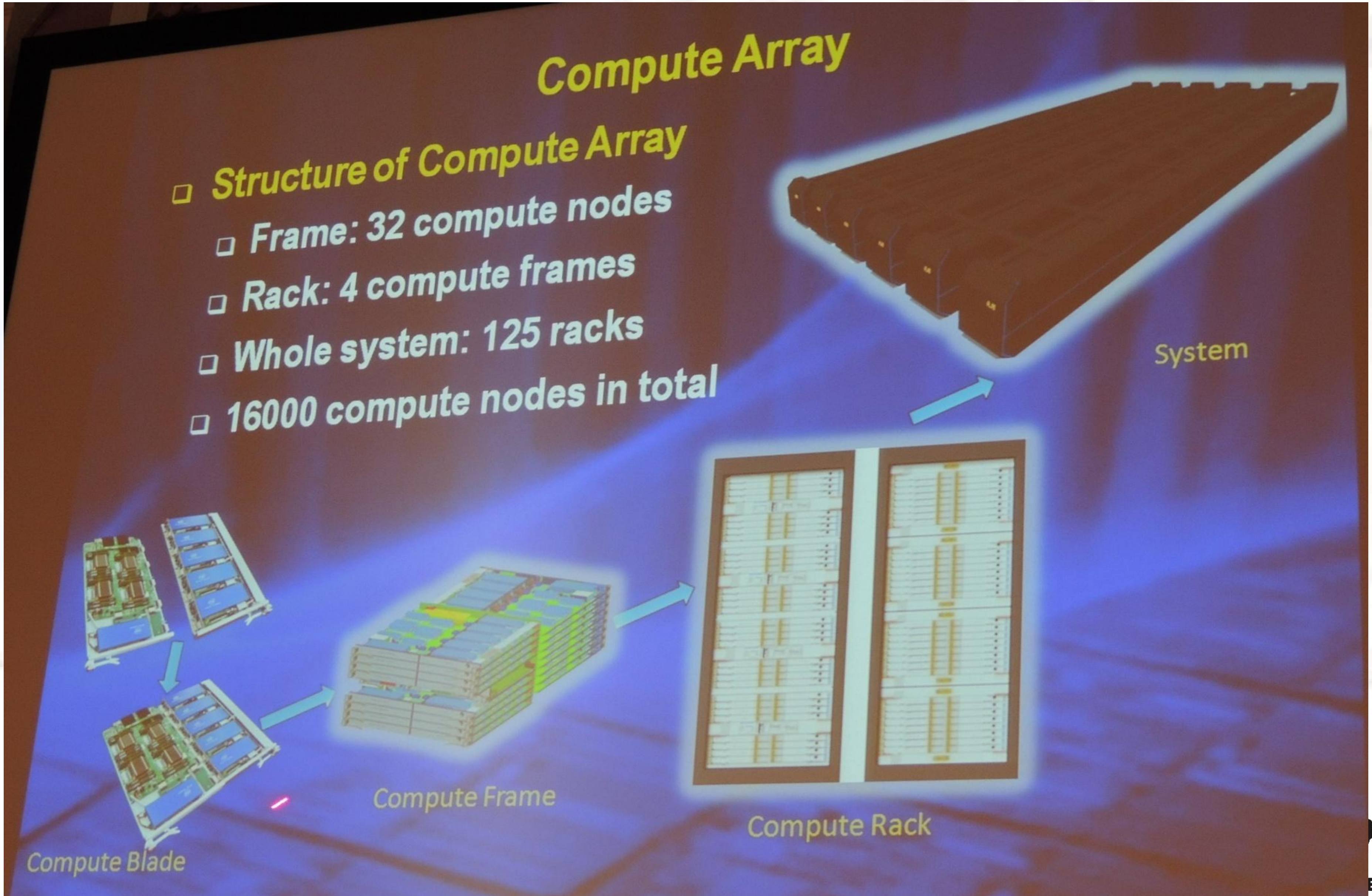
Comuter Node of TH-2



Computer Array of TH-2



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- *40960 Chinese-designed SW26010 manycore 64-bit RISC processors*
  X (256 processing cores + 4 auxiliary cores) per SW26010
  = *10,649,600 CPU cores*
- *The processing cores feature 64 KB of scratchpad memory for data (and 16 KB for instructions) and communicate via a network on a chip,*
- ***OS:*** *Sunway RaiseOS 2.0.5 on linux with its its own customized implementation of OpenACC2.0*

Fastest Supercomputer in the world

http://top500.org/ June. 2018

Rank	System	Cores	Rmax [TFlop/s]	Rpeak [TFlop/s]	Power [kW]
1	**Summit** - IBM Power System AC922, IBM POWER9 22C 3.07GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband , IBM DOE/SC/Oak Ridge National Laboratory United States	2,282,544	122,300.0	187,659.3	8,806
2	**Sunway TaihuLight** - Sunway MPP, Sunway SW26010 260C 1.45GHz, Sunway , NRCPC National Supercomputing Center in Wuxi China	10,649,600	93,014.6	125,435.9	15,371
3	**Sierra** - IBM Power System S922LC, IBM POWER9 22C 3.1GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband , IBM DOE/NNSA/LLNL United States	1,572,480	71,610.0	119,193.6	
4	**Tianhe-2A** - TH-IVB-FEP Cluster, Intel Xeon E5-2692v2 12C 2.2GHz, TH Express-2, Matrix-2000 , NUDT National Super Computer Center in Guangzhou China	4,981,760	61,444.5	100,678.7	18,482
5	**AI Bridg** Gold 614 National Japan			576.6	1,649
6	**Piz Daint** NVIDIA T Swiss Na Switzerl			326.3	2,272
7	**Titan** - C NVIDIA K20x , Cray Inc. DOE/SC/Oak Ridge National Laboratory United States			112.5	8,209
8	**Sequoia** - BlueGene/Q, Power BQC 16C 1.60 GHz, Custom , IBM DOE/NNSA/LLNL United States	1,572,864	17,173.2	20,132.7	7,890
9	**Trinity** - Cray XC40, Intel Xeon Phi 7250 68C 1.4GHz, Aries interconnect , Cray Inc. DOE/NNSA/LANL/SNL United States	979,968	14,137.3	43,902.6	3,844
10	**Cori** - Cray XC40, Intel Xeon Phi 7250 68C 1.4GHz, Aries interconnect , Cray Inc.	622,336	14,014.7	27,880.7	3,939

Summit”的计算能力比神威·太湖之光要快60%，比前美国超级计算机“Titan（泰坦）”要快8倍。

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Newest Top 10   June 2020

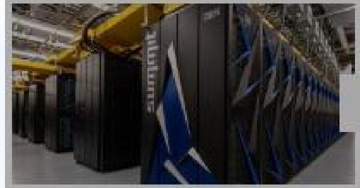
No.1 Fugaku(富岳)

日本 制造商：富士通

处理器核芯：7299072个；峰值(Rmax)：415530 TFlop/s

简介：

Fugaku超算原来被称为“Post K”，是曾经的世界第一K computer产品的第四代，采用ARM架构的富士通A64FX处理器，性能为第二名Summit的2.8倍。



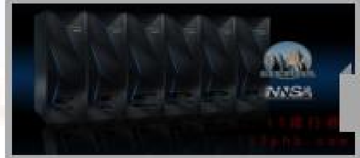
No.2 Summit（美国）

美国 制造商：IBM

处理器核芯：2414592个；峰值(Rmax)：148600 TFlop/s

简介：

顶点Summit是IBM和美国能源部橡树岭国家实验室（ORNL）推出的新超级计算机，Summit 要比神威·太湖之光快 60%，比同在橡树岭实验室的Titan——前美国超算记录保持者要快接近 8 倍。而在其之下，近 28,000 块英伟达 Volta GPU 提供了 95% 的算力。



No.3 Sierra（美国）

美国 制造商：IBM

处理器核芯：1572480个；峰值(Rmax)：94640 TFlop/s

简介：

Sierra超级计算机美国国家能源局橡树岭国家实验室已经给它定下来要做的事情，助力科学家在高能物理、材料发现、医疗保健等领域的研究探索。其中在癌症研究方面将用于名为“CANcer分布式学习环境（CANDLE）”的项目。

No.4 神威 太湖之光（Sunway TaihuLight）中国

中国 制造商：国家并行计算机工程技术研究中心

处理器核芯：10649600个；峰值(Rmax)：93015 TFlop/s

简介：

我国的神威 “太湖之光”超级计算机曾连续获得top500四届冠军，该系统全部使用中国自主知识产权的处理器芯片。

No.5 TH—2 天河二号（中国）

中国 制造商：国防科大

处理器核芯：4981760个；峰值(Rmax)：61445 TFlop/s

简介：

天河二号曾经6次蝉联冠军，采用麒麟操作系统，目前使用英特尔处理器，将来计划用国产处理器替换，不仅应用于助力探月工程、载人航天等政府科研项目，还在石油勘探、汽车飞机的设计制造、基因测序等民用方面大展身手。

No.6 HPC5（意大利）

意大利 制造商：DELL EMC

处理器核芯：669760个；峰值(Rmax)：35450 TFlop/s

简介：

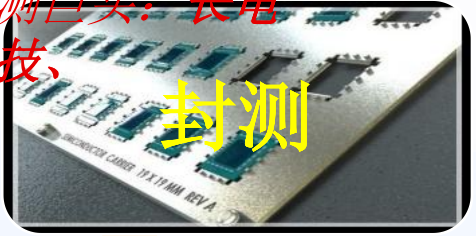
由DELL EMC公司为Eni能源公司打造的世界上功能最强大的工业用超级计算机，它的混合体系结构使分子模拟算法特别有效。

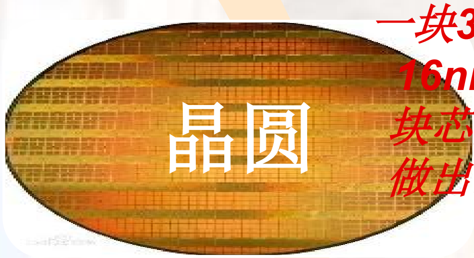
Chip technology

高纯硅要求99.999999999%,几乎全赖进口，传统霸主依然是德国Wacker和美国Hemlock(美日合资)

台湾的天下，排名世界第一的日月光，还跟着一堆实力不俗的小弟：矽品、力成、南茂、欣邦、京元电子。

大陆的三大封测巨头：长电科技、华天科技、通富微电

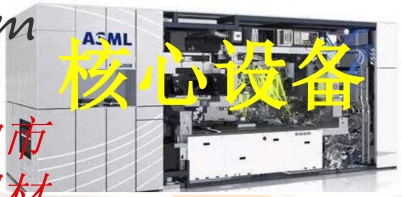




一块300mm直径的晶圆，16nm工艺可以做出100块芯片，10nm工艺可以做出210块芯片，价格可便宜了一半

刻蚀机，中国的状况要好很多，16nm已经量产，7-10nm也在上

离子注入机70%的市场份额是美国应用材料公司



光刻机，荷兰阿斯麦公司(ASML)唯一高端光刻机生产商（12，24，40台/年）

美英特尔、韩三星、日东芝、意法半导体；台湾的：旺宏电子；中国华润微电子等。

但中国高端芯片几乎空白

晶圆代工厂：台积电，中芯国际

高通、博通、AMD，中国台湾的联发科，大陆的华为海思

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What are the Big Bananas ?

Eckert-Mauchly Award

- http://www.computer.org/portal/web/awards/eckert
- **Administered jointly by ACM and IEEE Computer Society. The award of $5000 is given for contributions to computer and digital systems architecture where the field of computer architecture is considered at present to encompass the combined hardware-software design and analysis of computing and digital systems.**

Eckert-Mauchly Award Recipients

2018 Susan Eggers,
2017 Charles P. Thacker
2016 Uri Weiser
2015 Norman P. Jouppi
2014 Trevor Mudge
2013 James R. Goodman
2012 Algirdas Avizienis
2011 Gurindar (Guri) S. Sohi
2010 William J. Dally
2009 Joel S. Emer
2008 **Patterson, David**
2007 Valero, Mateo
2006 Pomerene, James H
2005 Colwell, Robert P.
2004 Brooks, Frederick P.
2003 Fisher, Joseph A. (Josh)
2002 Rau, B. Ramakrishna (Bob)
2001 **Hennessy, John**
2000 Davidson, Edward
1999 Smith, James E.
1998 Watanabe, T.
1997 **Tomasulo, Robert**
1996 **Patt, Yale**
1995 Crawford, John
1994 Thornton, James E.
1993 Kuck, David J
1992 **Flynn, Michael J.**
1991 Smith, Burton J.
1990 Batcher, Kenneth E.
1989 **Cray, Seymour**
1988 Siewiorek, Daniel P.
1987 **Amdahl, Gene M.**
1986 Cragon, Harvey G
1985 **Cocke, John**
1984 Dennis, Jack B.
1983 Kilburn, Tom
1982 **Bell, C. Gordon**
1981 Clark, Wesley A.
1980 Wilkes, Maurice V.
1979 Barton, Robert S.

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Big Men in Architecture(1)

- **2007Mateo Valero**

  http://personals.ac.upc.edu/mateo/

For important contributions to instruction level parallelism and superscalar processor design.

Big Men in Architecture(2)

- 2001 **Hennessy, John**

  For being the founder and chief architect of the MIPS Computer Systems and contributing to the development of the landmark MIPS R2000 microprocessor.

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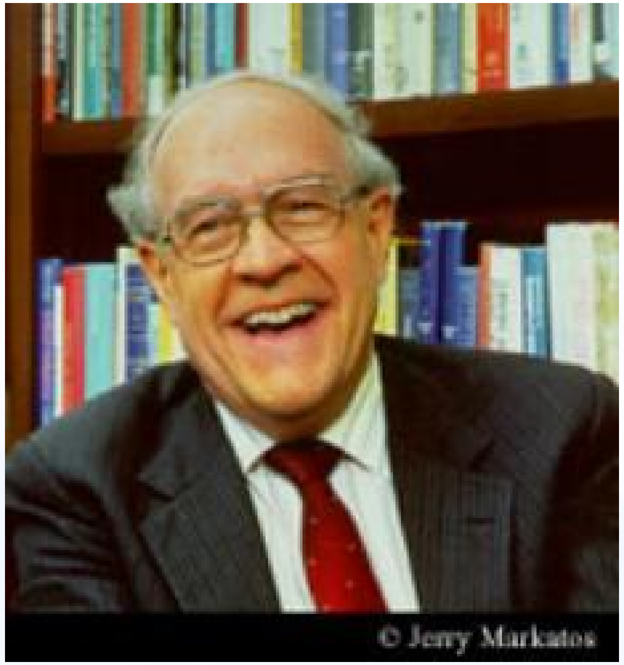
Big Men in Architecture(3)



Frederick P. Brooks

http://www.cs.unc.edu/~brooks/

2004 Eckert-Mauchly Award

"For the definition of computer architecture and contributions to the concept of computer families and to the principles of instruction set design; for seminal contributions in instruction sequencing, including interrupt systems and execute instructions; and for contributions to the IBM 360 instruction set architecture."

- **1999 ACM Turing Award**

  landmark contributions to computer architecture, operating systems, and software engineering."

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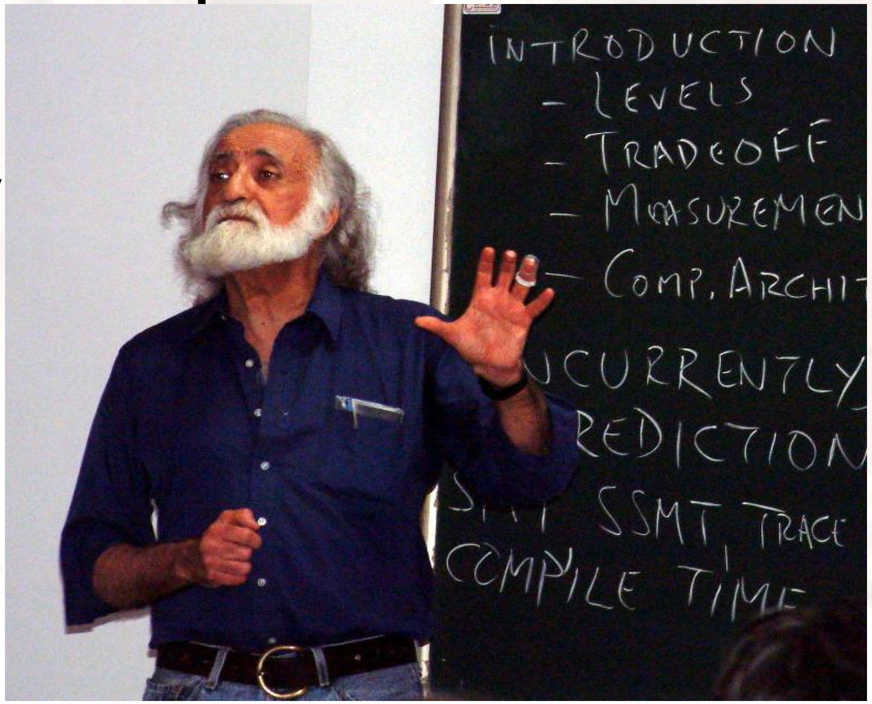
Big Men in Architecture(5)

- 1996 Yale Patt

  For important contributions to instruction level parallelism and superscalar processor design.

- Introduction to computing sy

  （2013-2017, 2019）



Big Men in Architecture(6)

- **1992 Michael J. Flynn**

  http://www.cpe.calpoly.edu/IAB/flynn.html

- For his important and seminal contributions to processor organization and classification, computer arithmetic and performance evaluation.

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Big Men in Architecture(7)

- 1989 Cray, Seymour
- For a career of achievements that have advanced supercomputing design.

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Big Men in Architecture(8)

From Computer Desktop Encyclopedia
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© 1999 Dr. Gene M. Amdahl

In 1975, Dr. Amdahl stands beside the Wisconsin Integrally Synchronized Computer (WISC), which he designed in 1950. It was built in 1952. (Image courtesy of Dr. Gene M. Amdahl.)

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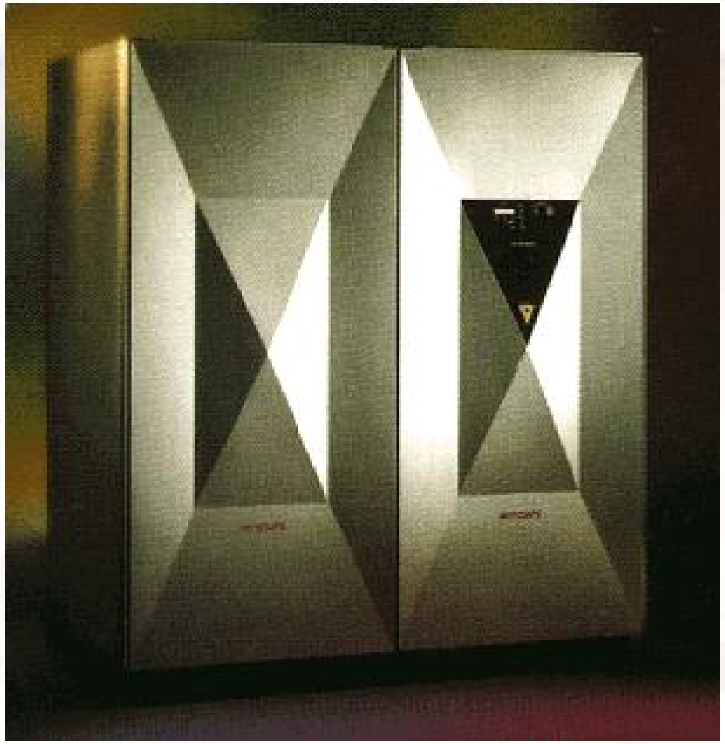

in computer architecture,
on look-ahead, and

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Top conferences and Journals

- **Top conference:**
  - ISCA
  - MICRO,
  - …
- **Top journals:**
  - IEEE Tran. on Computers — IF 3.131
  - ACM Tran. on Computer Systems — IF 1.917
  - IEEE Tran. on Parallel and Distributed Systems. — IF 3.402

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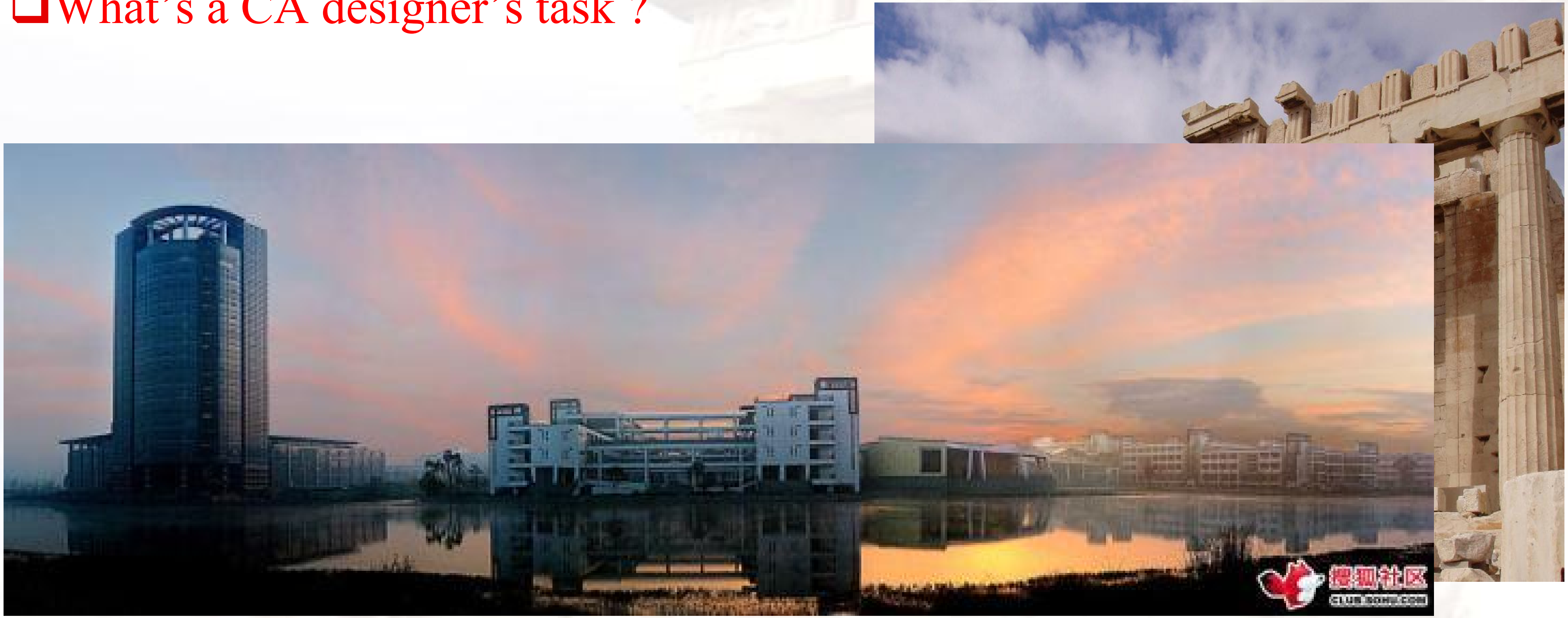
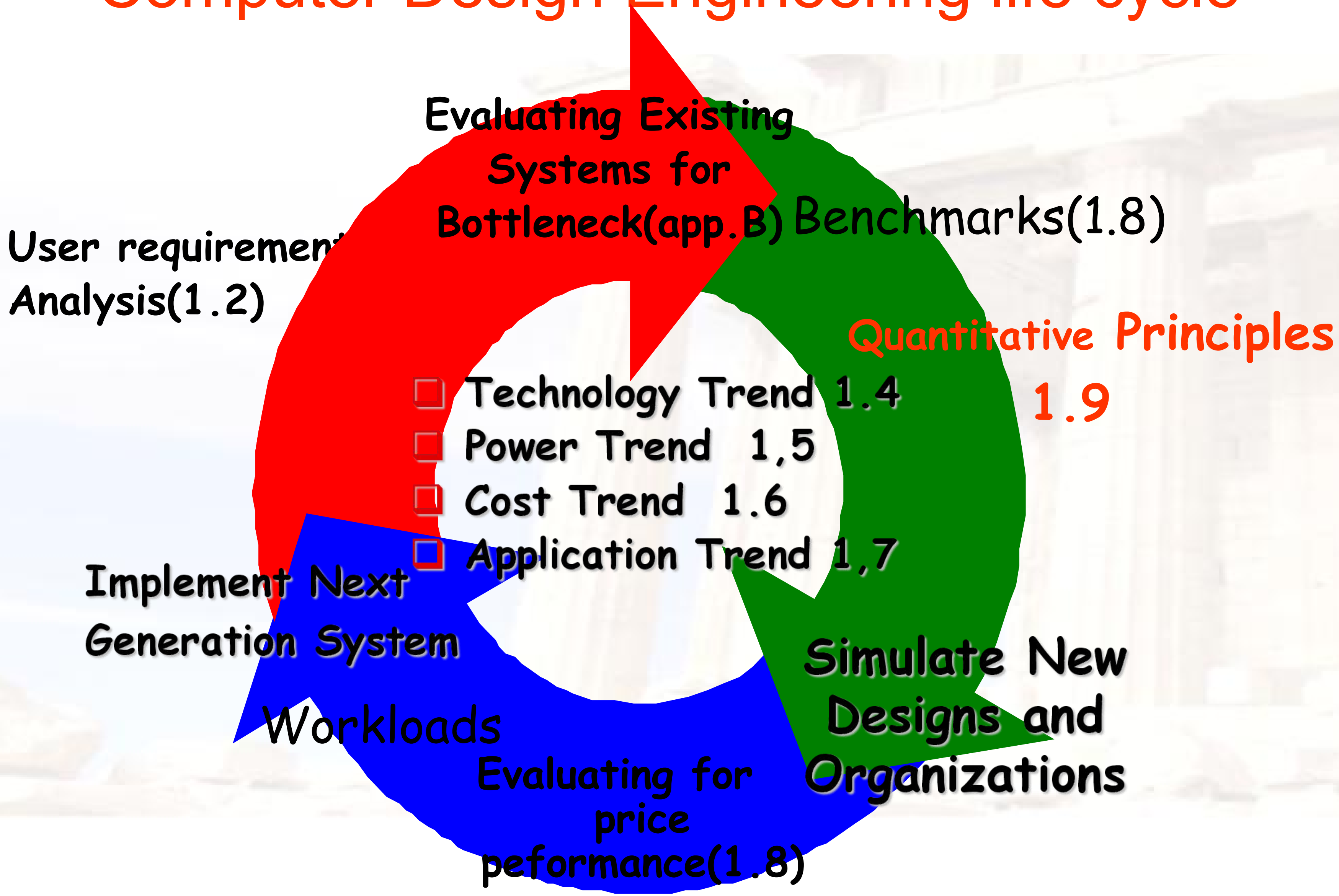
❑What's a CA designer's task ?



Computer Design Engineering life cycle



Topics in Chapter 1

1.1 Introduction
1.2 Classes of computers
1.3 Defining computer architecture and What's the task of computer design?
1.4 Trends in Technology
1.5 Trends in power in Integrated circuits
1.6 Trends in Cost
1.7 Dependability
1.8 Measuring, Reporting and summerizing Perf.
1.9 Quantitative Principles of computer Design
1.10 Putting it altogether

Summary: Task of computer design

- **Considerations**:
  - functional and non functional requirements
  - implementation complexity
    - Complex designs take longer to complete
    - Complex designs must provide higher performance to be competitive
  - **Technology trends**
    - Not only what's available today, but also what will be available when the system is ready to ship. (more on this later)
  - **Trends in Power in IC**
  - **Trends in cost**
- **Arguments**
  - Evaluate Existing Systems for Bottlenecks
- **Quantitative Principles**

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- End !
- Thank you !